SME Loan Origination System (LOS)

Business Analysis & Systems Analysis Document

Portfolio Case Study

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1. Document Control

Item	Detail
Document Title	SME Loan Origination System — Business & Systems Analysis
Version	1.0
Author	Abdullah Ibrahim Elsayed Mohamed Tawfik — Systems Analyst
Audience	Hiring Managers, Lead Analysts, Solution Architects, Fintech Recruiters
Purpose	Portfolio Case Study demonstrating end-to-end systems analysis
Confidentiality	Public (portfolio) — no real customer data used

Revision History

Version	Date	Author	Change
0.1	2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Initial draft
0.5	2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Added process models & requirements
1.0	2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Final — full document with diagrams & API spec

2. Executive Summary

This document presents a complete systems analysis for an SME Loan Origination System (LOS) for a mid-sized bank or lending institution. The system digitizes and controls the end-to-end process of originating a small-and-medium-enterprise loan — from application intake to disbursement — while enforcing credit policy, KYC/AML compliance, segregation of duties, and audit requirements.

Why this project matters as a portfolio piece

- It demonstrates the analyst's ability to model a single, complex, regulated business process with multiple human actors, external integrations, approval hierarchies, exception paths, and compliance constraints.
- It contrasts deliberately with consumer-facing projects by proving competence in enterprise, B2B, and regulated financial workflows.
- It shows reasoning behind the diagrams: why certain transitions are permitted, why certain data must be immutable, and why certain endpoints exist.

Core design stance: Depth over breadth. The scope is deliberately limited to Loan Origination — not loan servicing, collections, or full core banking — so that the analysis can go deep within one process rather than broad across many.

3. Business Context

3.1 Organizational Context

The client is a mid-sized commercial bank or non-bank lender that currently originates SME loans through a mixture of email, spreadsheets, shared folders, and a legacy core system. Loan officers manually collect documents, credit officers manually assess eligibility, and approvers sign off on paper or scanned PDFs. The result is slow turnaround, inconsistent decisions, poor auditability, and frequent regulatory findings.

3.2 Business Drivers

Driver	Description
Regulatory pressure	Central bank requires traceable credit decisions and KYC records
Competitive pressure	Fintechs originate SME loans in days; the bank takes weeks
Operational risk	Manual document handling causes missing files, expired IDs, duplicates
Cost of credit	Poor data quality increases default rates and rework
Customer experience	SMEs abandon applications when the process is opaque

3.3 Problem Statement

SME loan origination is slow, inconsistent, and poorly auditable because intake, KYC, credit assessment, approval, and disbursement are performed across disconnected tools with no enforced workflow, no single source of truth for application state, and no reliable audit trail.

4. Scope

4.1 In Scope

- Application intake (online + branch-assisted)
- Document upload & validation
- KYC / AML verification (via external provider)
- Credit bureau check (via external provider)
- Eligibility & scoring rules engine
- Manual credit review
- Multi-level approval workflow
- Offer generation & customer acceptance
- Disbursement request & confirmation
- Audit trail & regulatory reporting extracts

4.2 Out of Scope

- Loan servicing (repayments, statements)
- Collections & restructuring
- Core banking ledger (integration only)
- Accounting / GL entries

- Marketing / lead generation
- Mobile app (web + branch terminal only in this analysis)
- Full BPM engine selection (analysis is tool-agnostic)

4.3 Scope Boundary Rationale

The boundary is drawn around origination, not around the lender. This is a deliberate analytical decision: it clarifies which capabilities belong to the LOS and which are external systems (KYC provider, credit bureau, core banking). This boundary is what makes the DFD and API spec meaningful.

5. Stakeholder Analysis

Stakeholder	Type	Goal	Pain Today	Influence
SME Applicant	External	Fast, transparent decision	Opaque status, repeated document requests	Medium
Loan Officer	Internal	Submit complete applications quickly	Manual follow-ups	High
Credit Analyst	Internal	Assess risk consistently	Inconsistent data, no scoring support	High
Risk / Credit Committee	Internal	Enforce policy	No visibility into decision rationale	High
Compliance / AML Officer	Internal	Meet regulatory obligations	Fragmented KYC evidence	High
Operations	Internal	Disburse accurately	Re-keying between systems	Medium
Internal Audit	Internal	Verify decisions and controls	Incomplete audit trail	High
Regulator	External	Supervision	Late / inaccurate reporting	High
KYC Provider	External System	Verify identity	—	Medium
Credit Bureau	External System	Provide credit history	—	Medium
Core Banking	External System	Book the loan	—	High
IT / Security	Internal	Secure and maintain the system	—	Medium

5.1 Stakeholder Conflict & Resolution

A key tension: Loan Officers want speed, while Compliance and Risk want control. The analysis resolves this by automating deterministic checks (KYC, bureau, eligibility) so they do not block human review unnecessarily, by introducing parallel tasks (KYC + bureau run in parallel), by introducing SLA timers with escalation, and by making compliance evidence a by-product of the workflow rather than a separate manual step.

6. As-Is vs. To-Be

6.1 As-Is (Current State)

1. Applicant emails or visits branch; loan officer emails a PDF form.
2. Applicant returns form + scanned documents by email.
3. Loan officer saves documents into a shared folder and manually checks completeness.
4. Loan officer emails KYC documents to compliance; compliance replies by email.
5. Loan officer requests a credit bureau report manually via a portal.
6. Credit analyst opens a spreadsheet, manually scores, and writes a memo.
7. Memo is emailed to the credit committee; approvals are collected by email.
8. Offer is typed in Word, sent by email; applicant signs and returns a scan.
9. Operations re-keys the loan into the core banking system.
10. Audit requests evidence and reconstructs the file from emails.

As-Is pain points: no enforced workflow, no state model, no audit trail, duplicate entry, no SLA, no traceability, high operational risk.

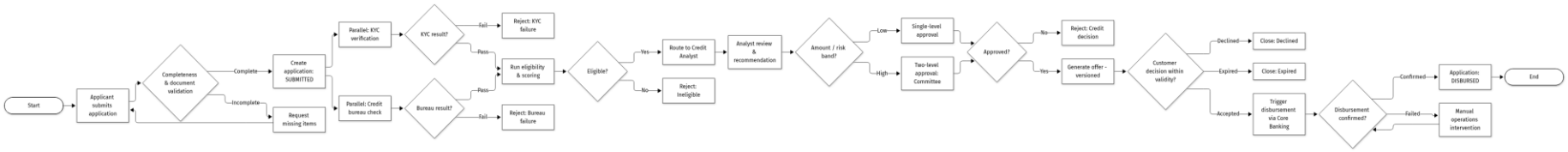
6.2 To-Be (Future State)

1. Applicant applies online or via branch terminal; system assigns a case ID and creates the application in state DRAFT.
2. System validates mandatory fields and document types; enforces SUBMITTED only when complete.
3. On submission, the system in parallel triggers KYC verification and credit bureau check via APIs.
4. Eligibility & scoring rules run automatically; deterministic rejections are returned with reasons.
5. If eligible, the application is routed to a Credit Analyst queue; the analyst reviews and records a recommendation.
6. Based on amount and risk band, the workflow routes to one or two approval levels (Credit Committee or delegated authority).
7. On approval, the system generates a versioned offer; the customer accepts or declines within a validity window.
8. On acceptance, Operations triggers disbursement via core banking integration; system records confirmation.
9. Every state change, decision, and document is written to an immutable audit trail.

To-Be improvements: enforced state machine, parallel checks, rule-based routing, immutable audit, SLA timers, single source of truth.

7. Business Process Model (BPMN-Style)

The following BPMN-style process is rendered from Mermaid.



7.1 Key Process Decisions

- Parallel KYC + Bureau: reduces turnaround by ~40% versus sequential execution.
- Deterministic rejection before human review: protects analyst time and makes outcomes consistent.
- Routing by amount & risk band: enforces delegated authority policy automatically.
- Offer validity window: prevents stale offers from being disbursed at outdated rates.
- Disbursement confirmation gate: ensures the LOS state matches the core banking reality.

8. Functional Requirements

ID	Requirement	Priority	Source
FR-01	System shall allow applicants to create and save a draft application.	Must	Applicant
FR-02	System shall enforce mandatory fields and required document types before submission.	Must	Loan Officer
FR-03	System shall verify applicant identity via the KYC provider API.	Must	Compliance
FR-04	System shall retrieve credit history via the credit bureau API.	Must	Credit Analyst
FR-05	System shall apply eligibility and scoring rules and store the result with a versioned rule-set ID.	Must	Risk
FR-06	System shall route applications to the correct approval level based on amount and risk band.	Must	Credit Committee
FR-07	System shall generate a versioned offer with validity window.	Must	Operations
FR-08	System shall record every state change in an immutable audit log.	Must	Audit
FR-09	System shall trigger disbursement to core banking and record confirmation.	Must	Operations
FR-10	System shall expose application status to the applicant.	Should	Applicant
FR-11	System shall escalate SLA breaches to supervisors.	Should	Operations
FR-12	System shall support re-submission after rejection with a linked new application.	Should	Loan Officer
FR-13	System shall allow parallel KYC and bureau checks.	Must	Analyst / Risk
FR-14	System shall prevent self-approval by the officer who submitted the application (segregation of duties).	Must	Compliance

FR-15	System shall produce regulatory reporting extracts.	Must	Regulator

9. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	KYC and bureau API calls shall return within 10s p95; the UI shall show a pending state, not block.
NFR-02	Availability	99.5% monthly availability during business hours.
NFR-03	Security	TLS 1.2+, encryption at rest (AES-256), secrets in vault.
NFR-04	Privacy	PII masked in logs; data retention per regulatory schedule.
NFR-05	Auditability	Audit log append-only; tamper-evident (hash chain).
NFR-06	Scalability	Support 500 concurrent applications, 50 concurrent analysts.
NFR-07	Usability	Branch terminal must be operable on low-bandwidth connections.
NFR-08	Interoperability	External integrations via REST/JSON with retries and idempotency keys.
NFR-09	Compliance	KYC/AML evidence retained per regulation; decisions explainable.
NFR-10	Maintainability	Rule sets are configurable without code deployment.
NFR-11	Recoverability	RPO ≤ 15 min, RTO ≤ 4 hours.
NFR-12	Localization	Support English and one local language for applicant UI.

10. Business Rules & Decision Tables

10.1 Eligibility Rules (BR)

Rule ID	Description
BR-01	Applicant age ≥ 21 and ≤ 65 at application date.
BR-02	Business registered ≥ 12 months.
BR-03	No active bankruptcy or fraud flag.
BR-04	Requested amount within product limit for the segment.
BR-05	Debt service coverage ratio ≥ threshold defined by product.
BR-06	KYC status = VERIFIED.
BR-07	Credit bureau score ≥ minimum for the product.

10.2 Approval Routing Decision Table

Requested Amount	Risk Band	Approval Path
≤ 50,000	Low	Single approver (Credit Analyst Lead)
≤ 50,000	Medium/High	Two-level (Analyst Lead + Committee)
50,001 – 250,000	Low	Two-level (Analyst Lead + Committee)
50,001 – 250,000	Medium/High	Two-level + Risk sign-off
Above 250,000	Any	Committee + Risk + Compliance sign-off

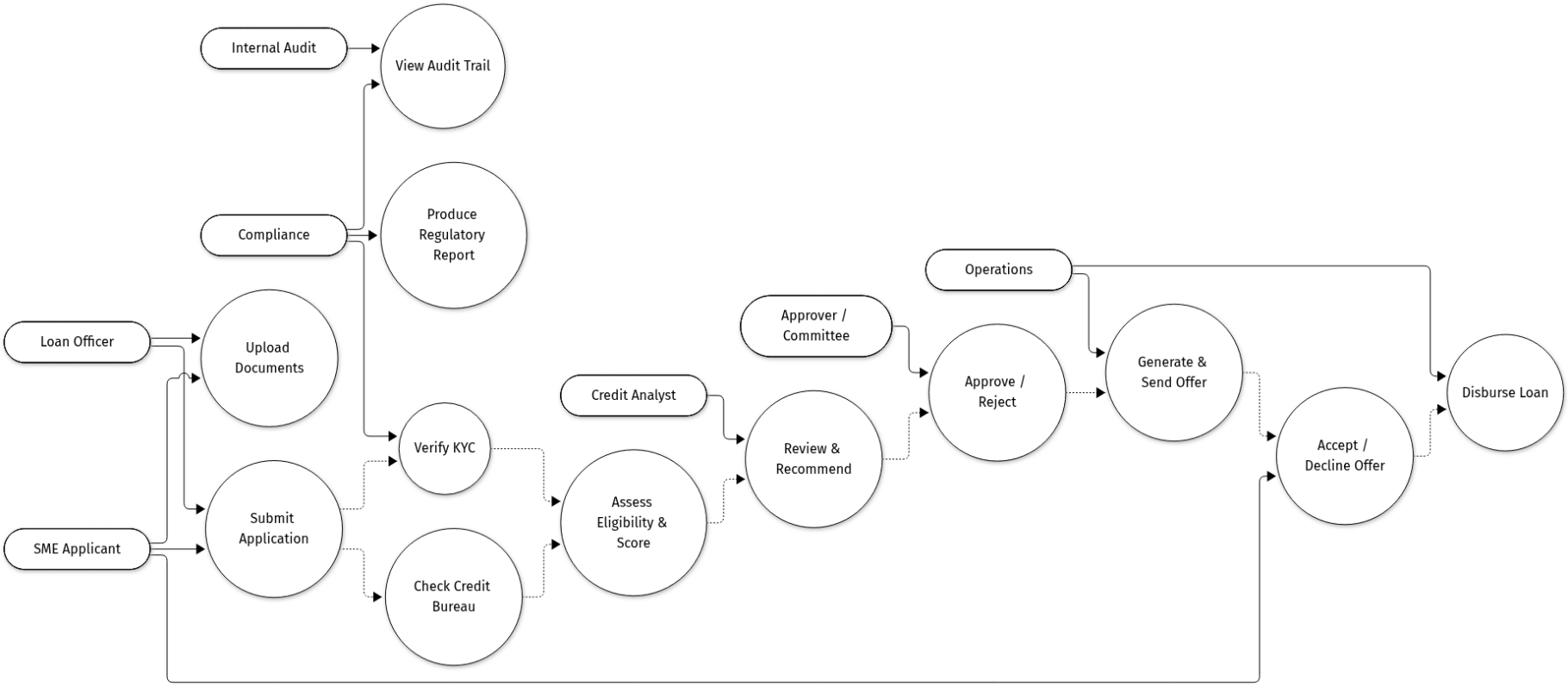
Analytical note: This table is the single source of truth for routing. It is externalized into a decision table (NFR-10) so policy changes do not require code releases.

10.3 Segregation of Duties (SoD) Rules

- The officer who submitted an application cannot approve it.
- The analyst who recommended cannot be the final approver for amounts above the single-approver threshold.
- Disbursement can only be triggered by Operations, not by the credit team.

11. Use Cases

11.1 Use Case Diagram



11.2 Primary Use Case — UC1: Submit Application

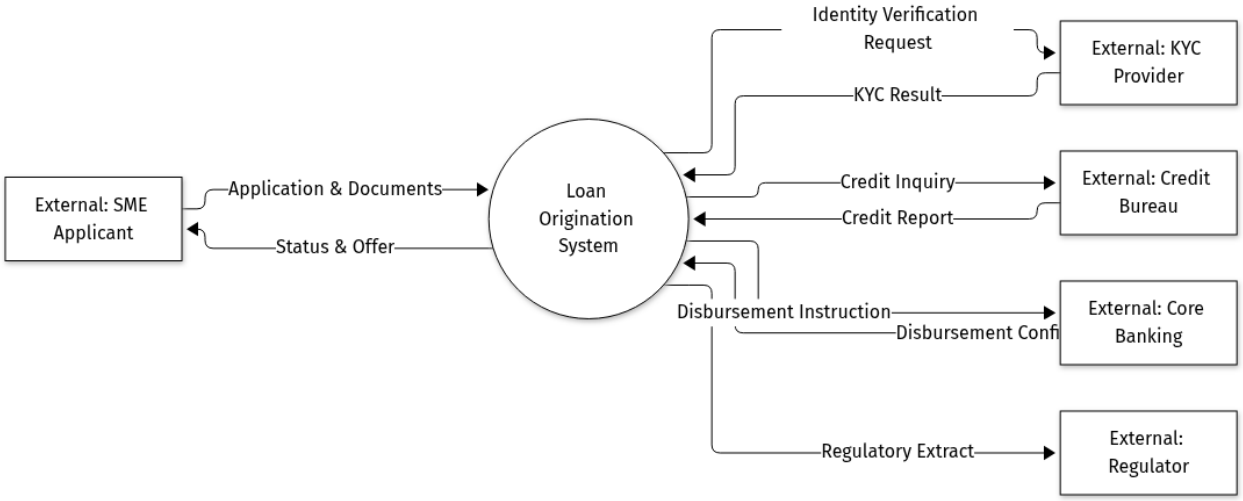
Field	Detail
Actor	SME Applicant / Loan Officer (on behalf)
Trigger	Applicant wants to apply for an SME loan
Preconditions	User is authenticated; applicant record exists or can be created
Main Flow	1. User starts application. 2. System creates draft. 3. User fills data. 4. User uploads required documents. 5. System validates completeness. 6. User submits. 7. System sets state to SUBMITTED and triggers KYC + bureau.
Alternate Flow	A1: Incomplete — system lists missing items and blocks submission.
Exception	E1: Duplicate active application detected — system links and warns.
Postconditions	Application in SUBMITTED; parallel checks initiated; audit entry written.

11.3 Secondary Use Case — UC7: Approve / Reject

Field	Detail
Actor	Approver / Credit Committee
Preconditions	Application in UNDER_REVIEW; analyst recommendation recorded; SoD satisfied
Main Flow	1. Approver opens application. 2. Reviews documents, score, recommendation. 3. Records decision + rationale. 4. System transitions state.
Alternate Flow	A1: Approver requests more info → state returns to INFO_REQUESTED.
Exception	E1: SoD violation → system blocks action.
Postconditions	State = APPROVED or REJECTED; decision rationale stored; audit entry written.

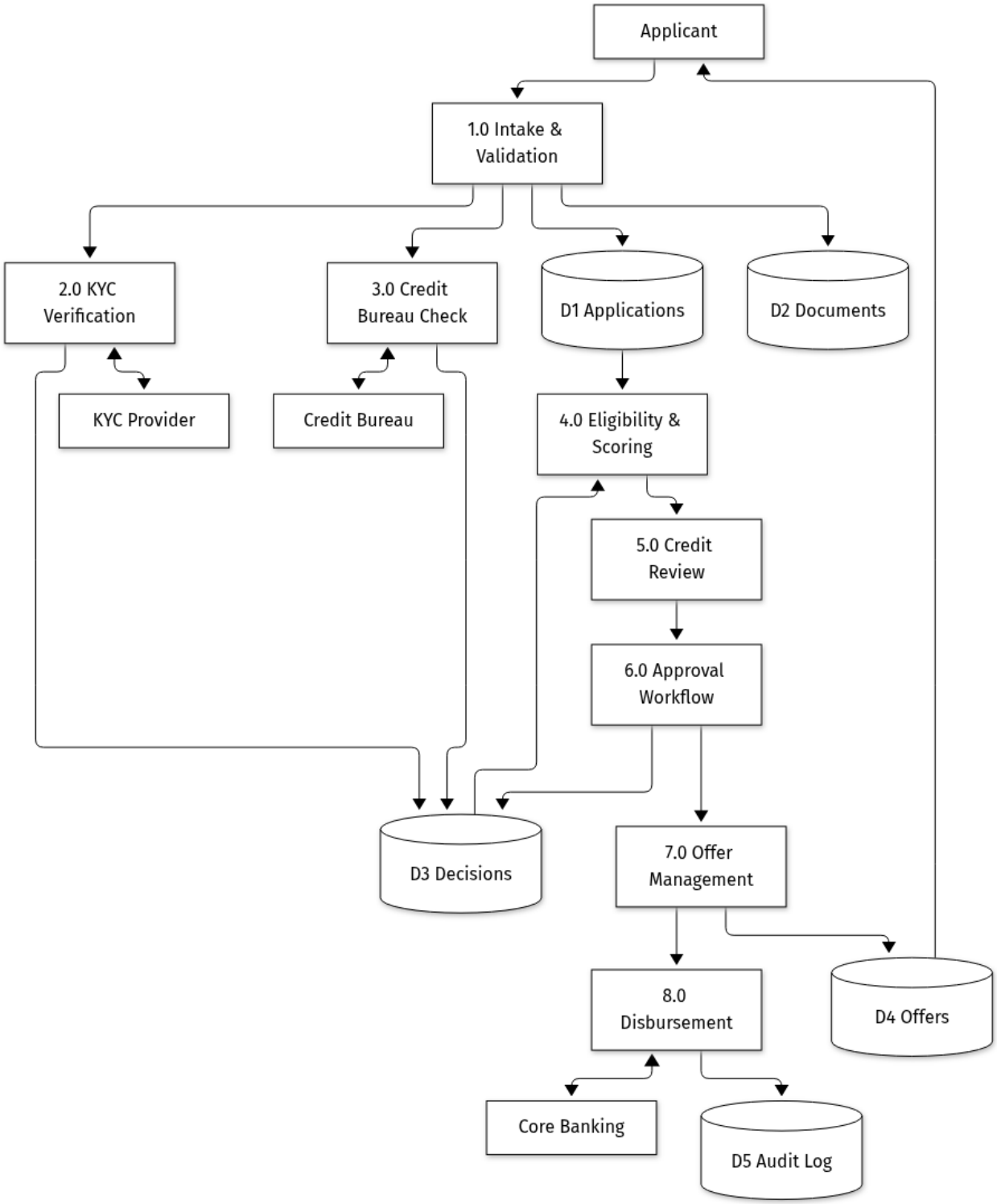
12. Analysis Diagrams

12.1 Context Diagram (DFD Level 0)

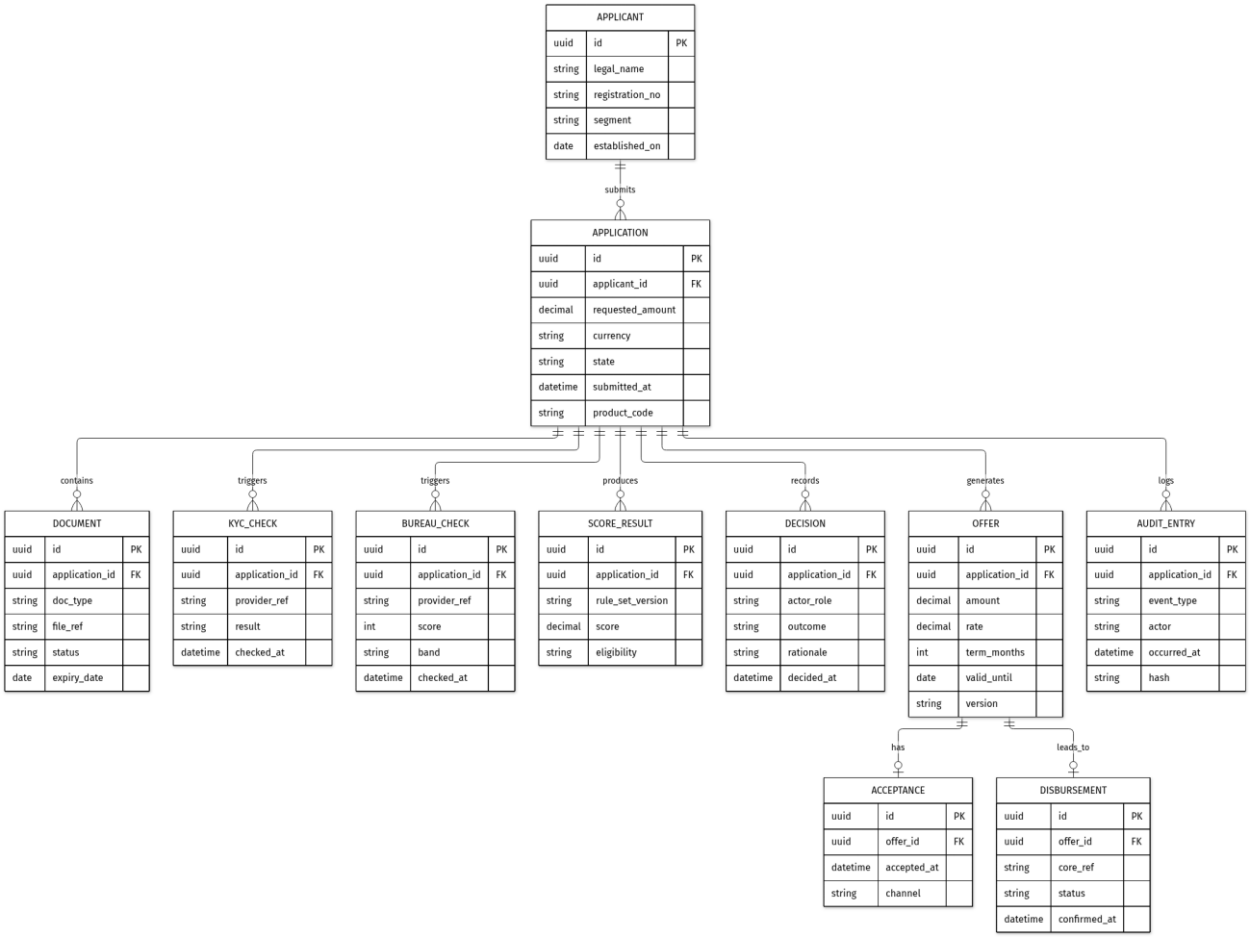


Analytical decision: External systems are modeled as terminators because they are outside the LOS boundary and interact via defined interfaces. The applicant is a single terminator even though a Loan Officer may act on their behalf — the analysis treats the officer as an internal actor (see Section 5).

12.2 DFD Level 1



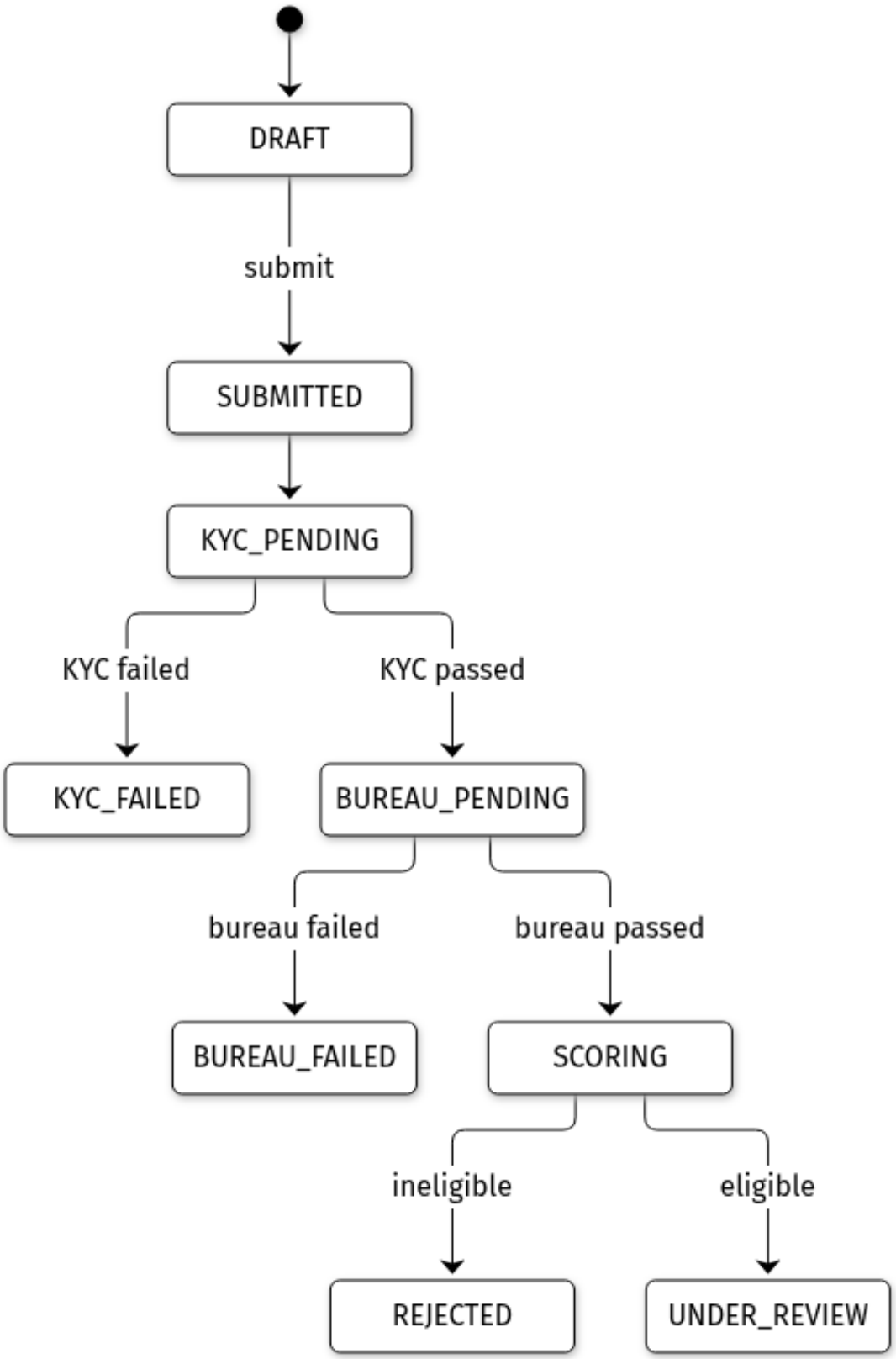
12.3 ERD (Data Model)



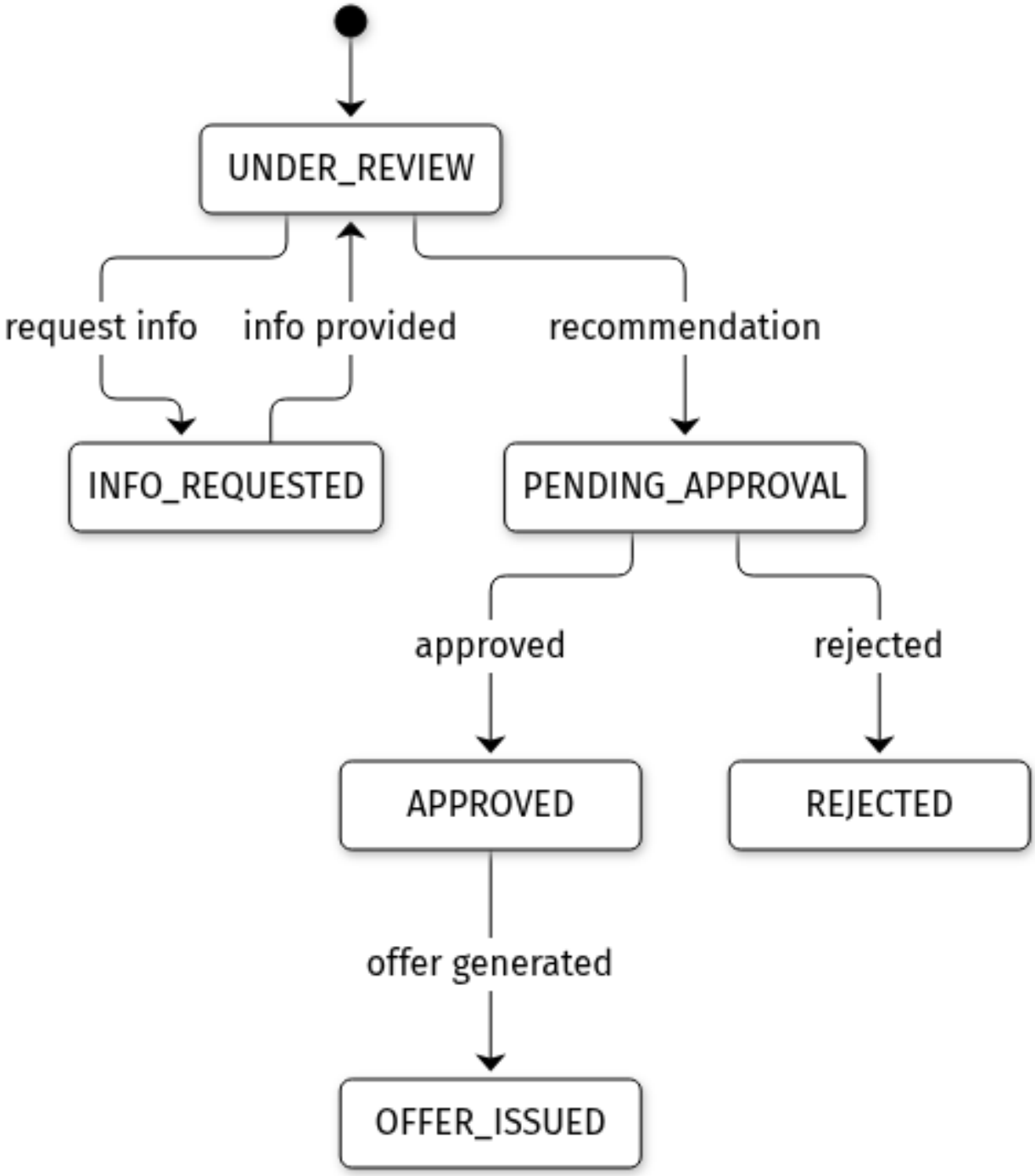
Analytical notes

- AUDIT_ENTRY has a hash column to support tamper-evidence (NFR-05).
- SCORE_RESULT.rule_set_version ties the decision to the exact policy version — required for regulatory explainability.
- OFFER.version supports re-issuance without mutating history.

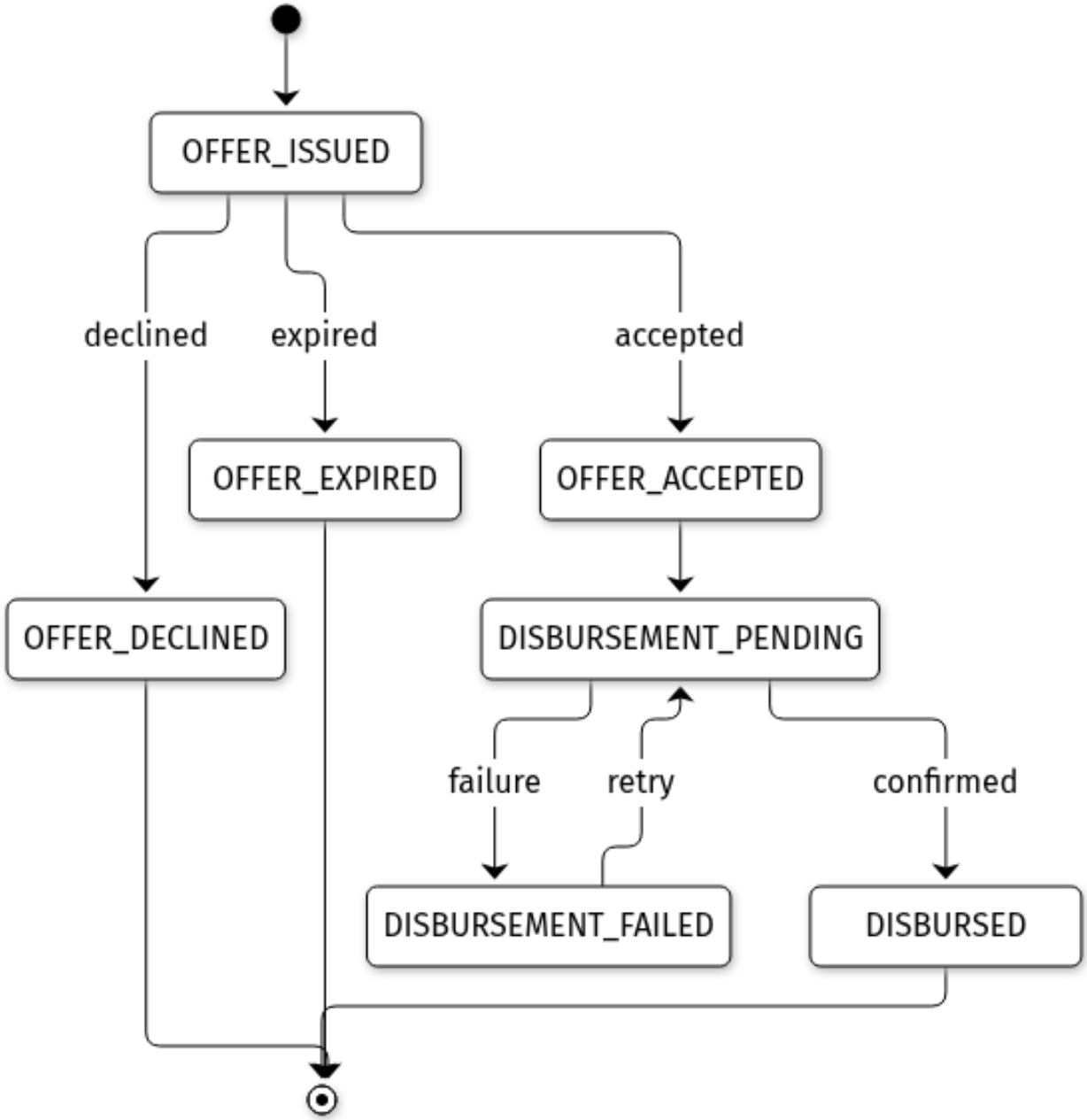
12.4 State Diagram — Application Lifecycle (Phase 1)



12.4 State Diagram — Application Lifecycle (Phase 2)



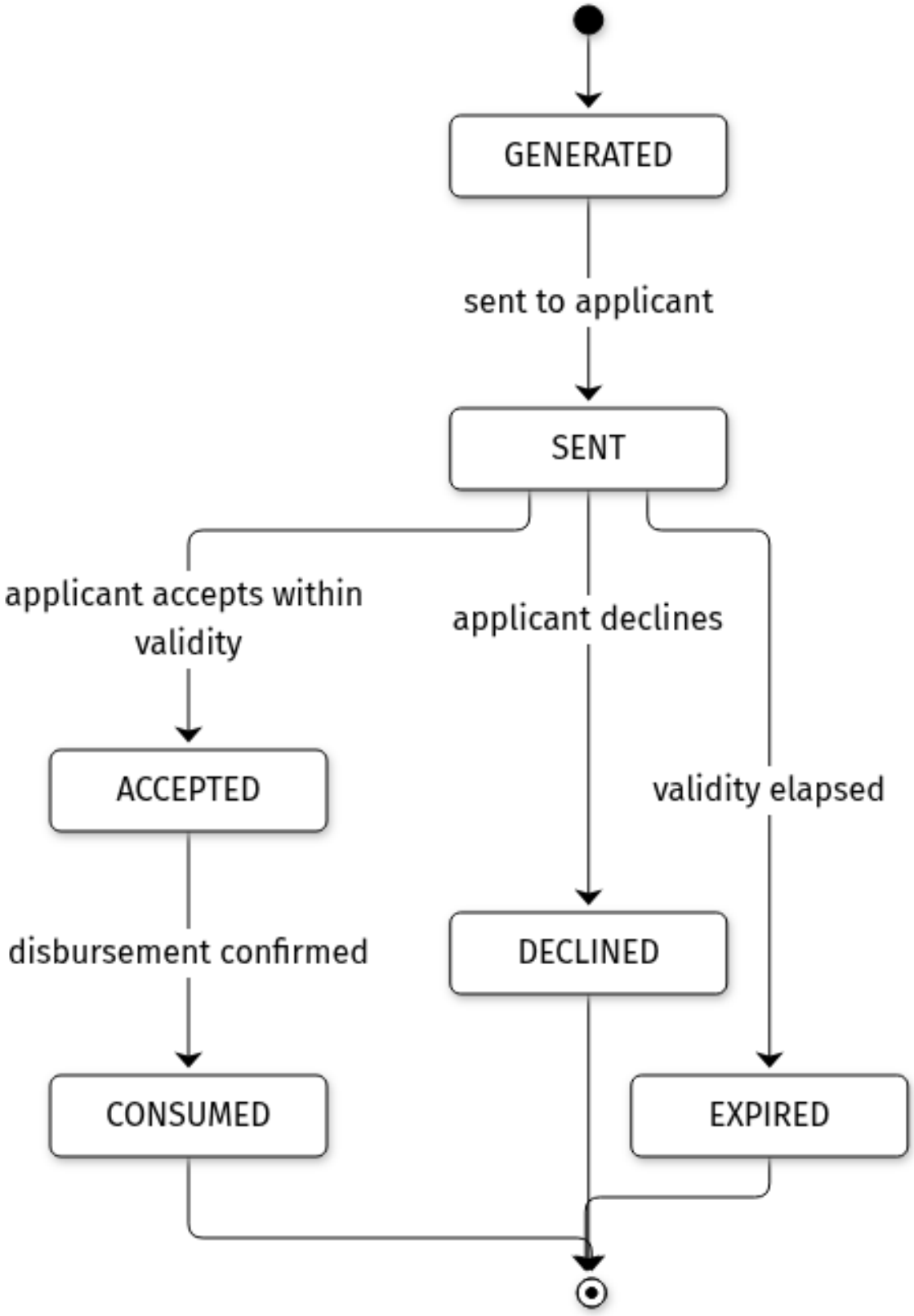
12.4 State Diagram — Application Lifecycle (Phase 3)



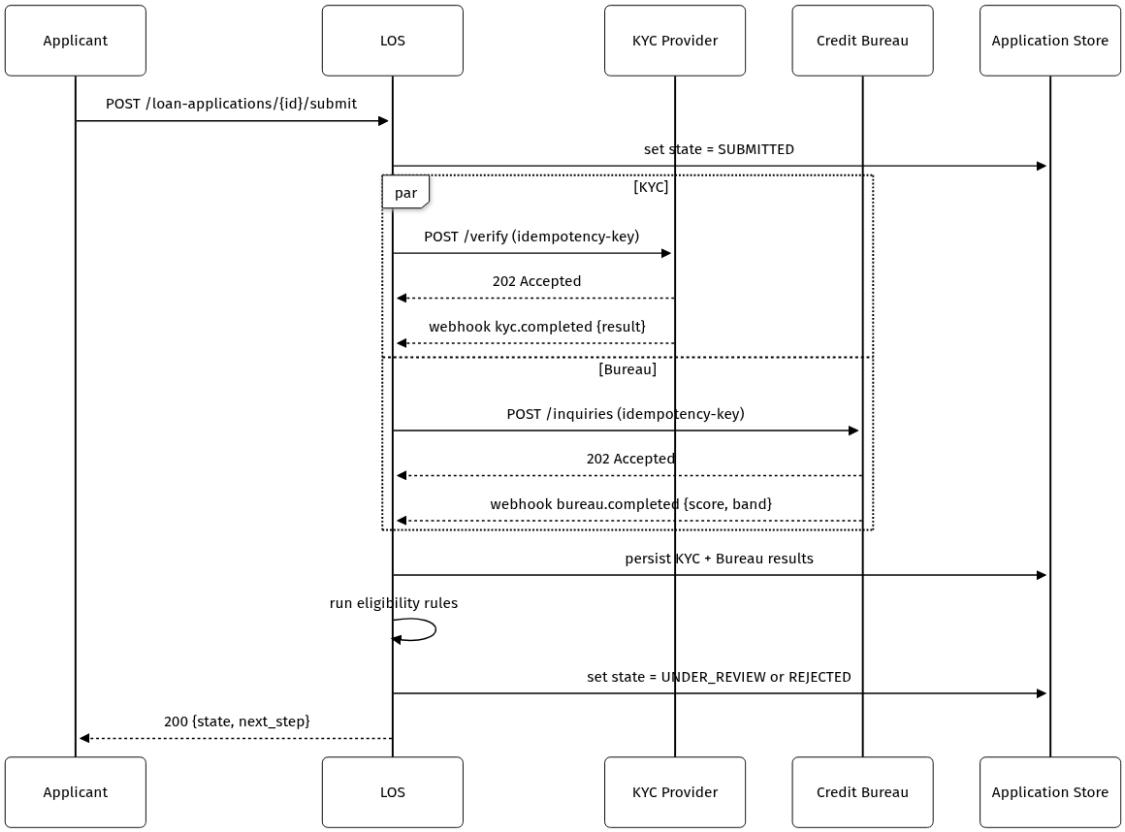
Analytical decisions behind the state machine

- No shortcut from SUBMITTED to APPROVED. Compliance requires KYC and bureau to be recorded first.
- INFO_REQUESTED is a bounce-back state, not a rejection, so information gaps do not corrupt decision history.
- OFFER_EXPIRED is a terminal state distinct from OFFER_DECLINED, because audit and regulatory treatment differ.
- DISBURSEMENT_FAILED is retryable, since core banking outages are transient.
- REJECTED cannot go back to PENDING_APPROVAL; re-application creates a new linked application (FR-12).

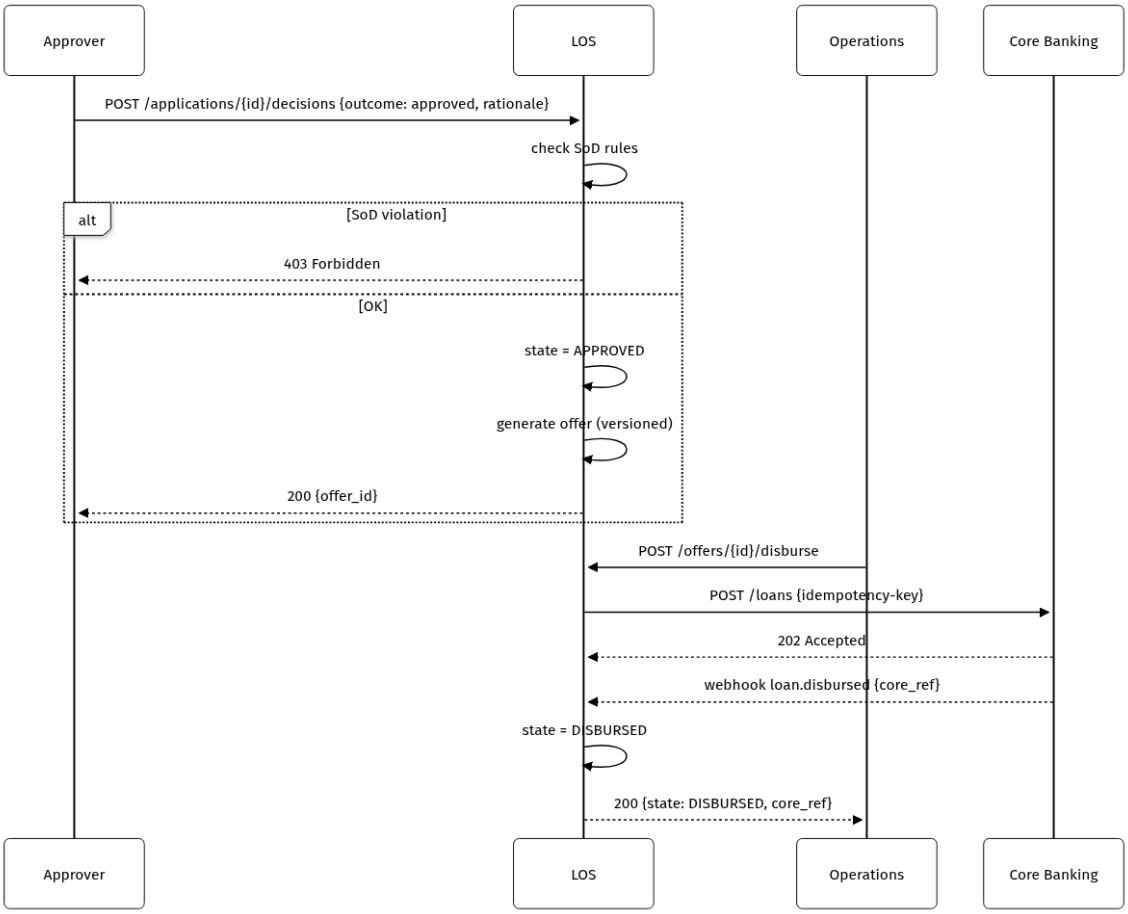
12.5 State Diagram — Offer Lifecycle



12.6 Sequence Diagram — KYC + Bureau (Parallel)



12.7 Sequence Diagram — Approval & Disbursement



13. API Specification (Selected Endpoints)

Style: REST/JSON, versioned via URL (/v1), idempotency via Idempotency-Key header on all mutating POSTs, authentication via OAuth2 client credentials (internal) and OIDC (external applicant portal).

13.1 POST /v1/loan-applications

Create a draft application.

Request

```
{
  "applicant_id": "0b6f...",
  "requested_amount": 150000,
  "currency": "EGP",
  "product_code": "SME_TERM_12M",
  "channel": "BRANCH"
}
```

Response

```
{
  "application_id": "app_7f3a...",
  "state": "DRAFT",
  "created_at": "2026-01-15T10:22:00Z"
}
```

Errors

400 invalid payload, 409 duplicate active application, 422 applicant not found.

13.2 POST /v1/loan-applications/{id}/submit

Submit application; triggers validation, then KYC + bureau in parallel.

Request

```
Headers: Idempotency-Key: <uuid>
```

Response

```
{
  "application_id": "app_7f3a...",
  "state": "SUBMITTED",
  "checks": ["KYC", "BUREAU"],
  "next_step": "AWAIT_CHECKS"
}
```

Errors

409 if state is not DRAFT; 422 if required documents missing.

13.3 GET /v1/loan-applications/{id}/status

Get current status and check results.

Request

```
(no body)
```

Response

```
{
  "application_id": "app_7f3a...",
  "state": "UNDER_REVIEW",
  "kyc": {"result": "PASSED", "checked_at": "..."},
  "bureau": {"score": 720, "band": "LOW", "checked_at": "..."},
  "scoring": {"rule_set_version": "v3.2", "eligibility": "ELIGIBLE"},
```

```
  "sla": {"due_at": "..."}
}
```

Errors

404 if not found.

13.4 POST /v1/loan-applications/{id}/decisions

Record a decision with SoD enforcement.

Request

```
{
  "outcome": "APPROVED",
  "rationale": "Meets all policy criteria; strong DSCR.",
  "approver_role": "COMMITTEE"
}
```

Response

```
{
  "application_id": "app_7f3a...",
  "state": "APPROVED",
  "decision_id": "dec_22a1...",
  "soD_checked": true
}
```

Errors

403 SoD violation; 409 state not PENDING_APPROVAL.

13.5 POST /v1/offers/{id}/accept

Customer accepts offer.

Request

```
{
  "channel": "ONLINE",
  "accepted_at": "2026-01-20T14:00:00Z"
}
```

Response

```
{
  "offer_id": "off_551c...",
  "state": "ACCEPTED"
}
```

Errors

410 offer expired; 409 already declined.

13.6 POST /v1/offers/{id}/disburse

Operations triggers disbursement.

Request

```
Headers: Idempotency-Key: <uuid>
{
  "account_id": "acc_9982...",
  "value_date": "2026-01-22"
}
```

Response

```
{
  "disbursement_id": "disb_0031...",
  "state": "PENDING"
}
```

Errors

403 role not Operations; 409 state not OFFER_ACCEPTED.

13.7 GET /v1/audit/applications/{id}

Response (paginated)

```
{
  "entries": [
    {"event": "STATE_CHANGED", "from": "DRAFT", "to": "SUBMITTED", "actor": "user_104", "at":
"..."},
    {"event": "DECISION_RECORDED", "outcome": "APPROVED", "actor": "user_207", "at": "..."}
  ]
}
```

13.8 Inbound Webhooks

Event	Source	Purpose
kyc.completed	KYC Provider	Update KYC result
kyc.failed	KYC Provider	Trigger rejection path
bureau.completed	Credit Bureau	Update score
loan.disbursed	Core Banking	Confirm disbursement

Analytical note: All inbound webhooks must be idempotent (event ID deduplication) and signature-verified. This is captured in NFR-08 and NFR-03.

14. RBAC Matrix

Capability	Applicant	Loan Officer	Credit Analyst	Approver	Ops	Compliance	Auditor	Admin
Create draft application	✓	✓	—	—	—	—	—	—
Upload documents	✓	✓	—	—	—	—	—	—
Submit application	✓	✓	—	—	—	—	—	—
View KYC result	—	✓	✓	✓	—	✓	✓	✓
Record recommendation	—	—	✓	—	—	—	—	—
Record decision	—	—	—	✓	—	—	—	—
Generate offer	—	—	—	—	✓	—	—	—
Trigger disbursement	—	—	—	—	✓	—	—	—
View audit trail	—	—	—	—	—	✓	✓	✓
Configure rules	—	—	—	—	—	✓	—	✓
Produce regulatory report	—	—	—	—	—	✓	✓	✓

15. Audit & Compliance

15.1 Audit Event Schema

Every state transition, decision, and document action produces an audit entry with:

- event_id, application_id
- event_type (STATE_CHANGED, DECISION_RECORDED, DOCUMENT_UPLOADED, OFFER_ISSUED, DISBURSEMENT_CONFIRMED)
- actor (user ID + role)
- occurred_at (UTC)
- payload (diff of before/after)
- prev_hash + hash (tamper-evident chain)

15.2 Regulatory Traceability

Every decision must resolve to:

- The exact rule-set version applied (from SCORE_RESULT.rule_set_version).
- The evidence (KYC result, bureau score, documents) used.
- The actor and timestamp of the decision.

16. Risk Analysis

ID	Risk	Likelihood	Impact	Mitigation
R-01	KYC provider outage blocks intake	Medium	High	Queue submissions; retry with idempotency; degrade to manual KYC pending
R-02	Bureau returns stale data	Low	High	Record provider_ref + timestamp; treat >30 days as invalid
R-03	SoD bypass by admin	Low	High	SoD rules enforced in workflow engine; admin actions audit-logged
R-04	Offer disbursed at outdated rate	Medium	High	Validity window; re-check before disbursement
R-05	Duplicate applications	Medium	Medium	Duplicate detection on submit (same applicant + product + window)
R-06	Audit log tampering	Low	High	Hash chain + append-only storage + periodic external anchoring
R-07	Rule misconfiguration	Medium	High	Versioned rule sets; 4-eyes review for rule changes
R-08	PII leakage in logs	Medium	High	Log masking; no PII in error payloads

17. Assumptions & Constraints

17.1 Assumptions

- KYC and credit bureau providers expose REST APIs with webhooks and idempotency support.
- Core banking supports a disbursement API and returns a confirmation.
- Regulatory requirements include explainable decisions and ≥ 7-year retention.
- Users are authenticated via the bank's identity provider (OIDC).

17.2 Constraints

- Must integrate with the existing core banking (no replacement in scope).
- Must run on the bank's on-prem or approved cloud environment.
- Must comply with local data residency rules.
- Must be operable by branch staff on low-bandwidth links.

18. KPIs & SLAs

KPI	Target
Median time from SUBMITTED → decision	≤ 3 business days
KYC + bureau completion (p95)	≤ 10 seconds
Disbursement confirmation (p95)	≤ 1 business day after acceptance
Applications rejected due to missing docs after submission	< 2%
SoD violations blocked by system	100%
Audit trail completeness	100%
SLA breach notifications sent	100% within 1 hour

19. Traceability Matrix (Sample)

Requirement	Use Case	Diagram	API
FR-03 KYC	UC3	12.1, 12.6	13.2, 13.8
FR-04 Bureau	UC4	12.1, 12.6	13.2, 13.8
FR-05 Scoring	UC5	12.2, 12.4	13.3
FR-06 Routing	UC7	12.4, 10.2	13.4
FR-07 Offer	UC8	12.5	13.5
FR-08 Audit	UC11	12.3, 15.1	13.7
FR-09 Disbursement	UC10	12.7	13.6, 13.8
FR-14 SoD	UC7	12.4, 14	13.4

20. "What I Would Improve" (Self-Critique)

- BPMN over flowchart? In production, this process would be modeled in true BPMN 2.0 (with gateways, message flows, pools). The Mermaid flowchart here conveys the logic clearly for a portfolio, but I would migrate it to BPMN 2.0 in a real engagement.
- Rules engine depth? The decision table is illustrative. A real engagement would define scoring weights, thresholds, and override procedures with the Risk team.
- Non-functional depth? NFRs are stated as targets; in production they would be validated with load tests and threat modeling.
- Regulatory variability? Requirements differ per jurisdiction; I scoped to generic "explainable decision + retention" and would localize in real delivery.

21. Glossary

Term	Meaning
LOS	Loan Origination System
KYC	Know Your Customer
AML	Anti-Money Laundering
SoD	Segregation of Duties
DSCR	Debt Service Coverage Ratio
BPMN	Business Process Model and Notation
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
RBAC	Role-Based Access Control
SLA	Service Level Agreement

RPO / RTO	Recovery Point / Time Objective

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